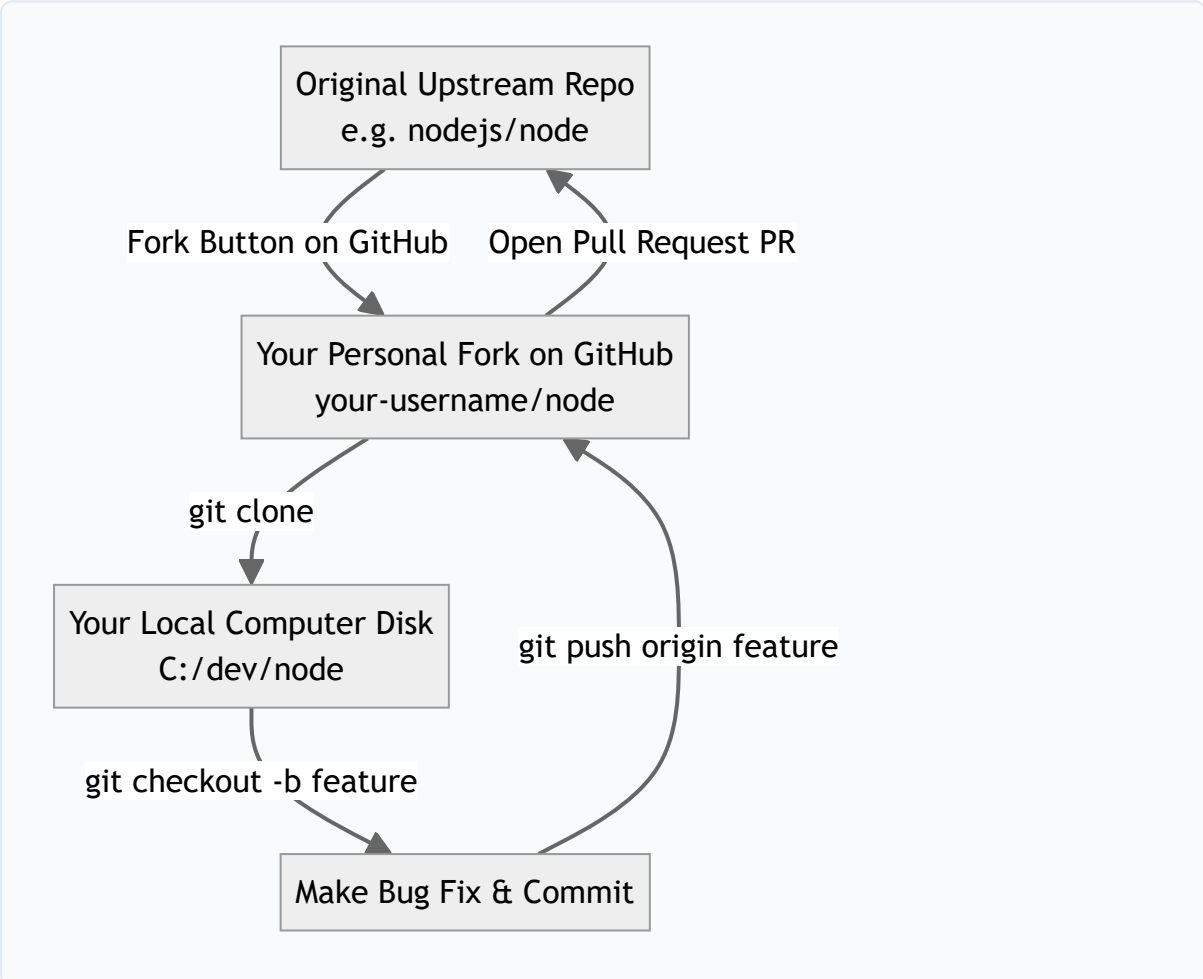
Submodule 35.20: What is Forking & Open Source Contribution

1. What is Forking?

A **Fork** is a complete, independent copy of another user's or organization's GitHub repository created directly inside your personal GitHub account on GitHub's cloud servers.

Fork vs Clone:

- **Clone**: Copies a repository from GitHub down to your local physical computer (`github.com` o Local Machine).
- **Fork**: Copies a repository from someone else's GitHub account to your personal GitHub account (`github.com/upstream/repo` o `github.com/your-username/repo`).



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2. Why Fork a Repository?

1. **Lack of Direct Write Access**: Open-source repositories (e.g., PyTorch, React, Linux) do not grant direct write/push access to random developers. Forking allows you to experiment freely in your own copy.
2. **Proposing Contributions (Pull Requests)**: Once you implement a bug fix or feature in your fork, you can open a PR to suggest merging your changes back into the original project.
3. **Starting a Derivative Project**: Taking an existing open-source project and expanding it into a new independent product.

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3. The 6-Step Open Source Contribution Protocol

1. **Fork**: Navigate to the target repository on GitHub and click the **Fork** button (top-right).
2. **Clone Your Fork**:

```
git clone https://github.com/your-username/repo.git
cd repo
```

3. **Create Feature Branch**:

```
git checkout -b fix-memory-leak
```

4. **Implement & Commit**:

```
git add .
git commit -m "Resolve memory leak in buffer allocation"
```

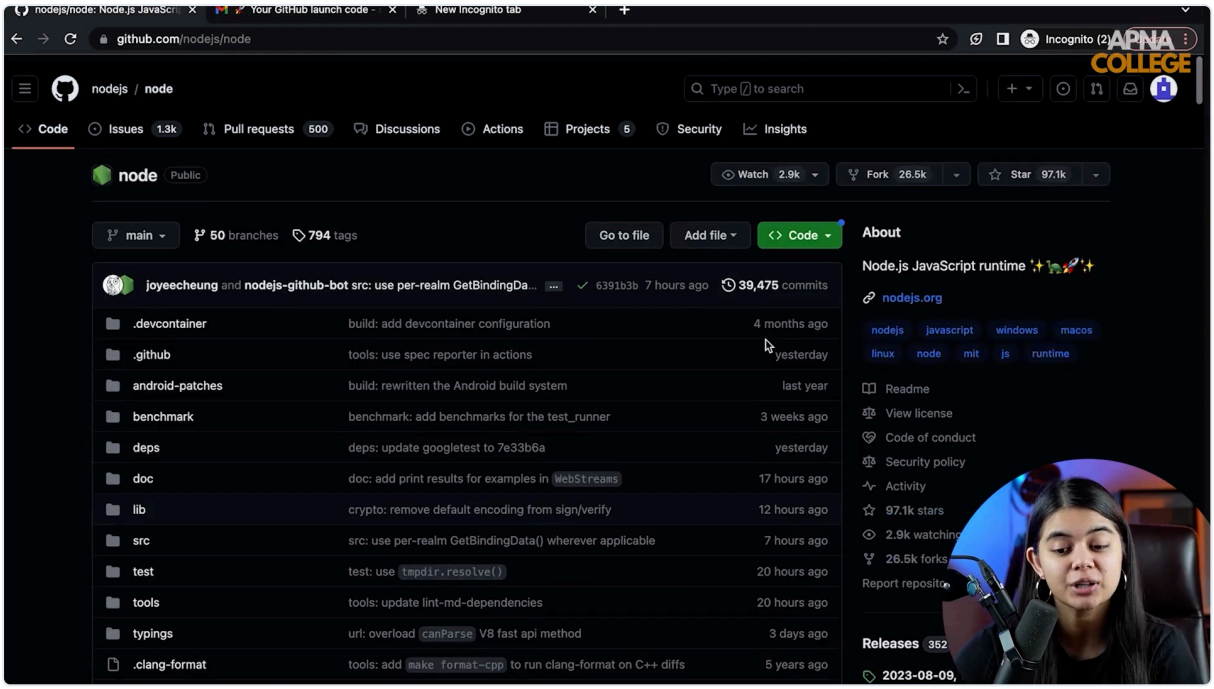
5. **Push to Your Fork**:

```
git push origin fix-memory-leak
```

6. **Open Pull Request**:
    - Go to the original upstream repository on GitHub.
    - Click **New Pull Request** *o* compare across forks *o* submit for project maintainer review.

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4. Visual Reference


github.com/nodejs/node
nodejs / node
Code
Issues 1.3k
Pull requests 500
Discussions
Actions
Projects 5
Security
Insights
node Public
Watch 2.9k
Fork 26.5k
Star 97.1k
main
50 branches
794 tags
Go to file
Add file
Code
joyeecheung and nodejs-github-bot src: use per-realm GetBindingDa... 6391b3b 7 hours ago 39,475 commits
.devcontainer build: add devcontainer configuration 4 months ago
.github tools: use spec reporter in actions yesterday
android-patches build: rewritten the Android build system last year
benchmark benchmark: add benchmarks for the test_runner 3 weeks ago
deps deps: update googletest to 7e33b6a yesterday
doc doc: add print results for examples in WebStreams 17 hours ago
lib crypto: remove default encoding from sign/verify 12 hours ago
src src: use per-realm GetBindingData() wherever applicable 7 hours ago
test test: use tmpdir.resolve() 20 hours ago
tools tools: update lint-md-dependencies 20 hours ago
typings url: overload canParse V8 fast api method 3 days ago
.clang-format tools: add make format-cpp to run clang-format on C++ diffs 5 years ago
About
Node.js JavaScript runtime
nodejs.org
nodejs javascript windows macos linux node mit js runtime
Readme
View license
Code of conduct
Security policy
Activity
97.1k stars
2.9k watching
26.5k forks
Releases 352
2023-08-09,
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